

5 Interesting Reference Designs FOR DALI-BASED LED LIGHTING

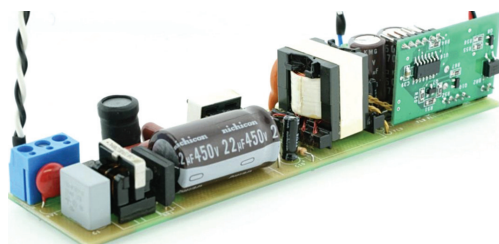
These DALI reference designs enable quick control, dimming, and testing of LED lighting without additional hardware or complex setup.

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DALI (digital addressable lighting interface) has become the standard for reliable communication and control in modern LED lighting systems, enabling functions like switching, dimming, and status reporting across multiple devices. For engineers developing DALI-based lighting, reference designs provide ready-made hardware and software solutions that simplify integration, reduce development time, and ensure DALI protocol compliance. These designs address key needs such as smooth dimming, two-way communication, electrical isolation, and stable LED current, while supporting configurations ranging from single LED drivers to multi-device networks.

LED driver with DALI dimming

This reference design enables you to build LED drivers with integrated DALI dimming, without adding extra control or power stages. It supports two-way DALI communication,



allowing the driver to receive commands and report status while keeping the control side electrically isolated. LED brightness is adjusted smoothly from 0 to 100 per cent using PWM-based current control, preventing flicker and visible dimming steps. The design keeps DALI inactive at startup to ensure stable turn-on, then enables dimming once normal operation begins. A regulated 5V supply powers the DALI microcontroller, and simple optocoupler isolation prevents ground loops. By handling DALI communication,

dimming control, isolation, and power within the reference design, it reduces firmware effort, the need for extra circuitry, and integration risk for engineers building DALI-compatible LED lighting systems.

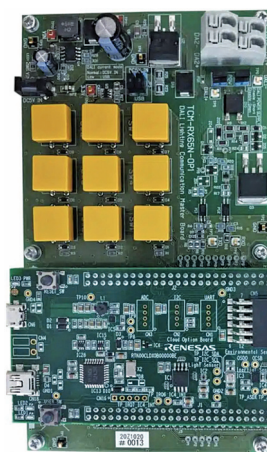
Applications. DALI-controlled LED lighting systems that require reliable, smooth dimming, and two-way communication.

Link. <https://www.electronicsforu.com/electronics-projects/led-driver-with-dali-dimming-reference-design>

OEM Brand. Power Integration

DALI-2 lighting system

Design engineers working with DALI often face challenges in understanding the protocol, connecting multiple luminaires, and testing dimming behaviour without building a complete system. This DALI lighting control solution addresses those challenges by providing a ready reference setup that combines hardware, software, and basic cloud support. It allows one controller to manage up to 64 DALI devices, making it easier to test dimming, communication, and system behaviour in real DALI networks. A built-in DALI-2 interface removes the need to design and debug the protocol from scratch. You can control lights through a PC, over the web, or using simple switches, depending on what you need to test. This helps you test DALI features such as group control, scenes, and response timing under real-world conditions.



Applications. The solution simplifies